

RAFT and Two Memories

DATA 443/643 · Week 10, note 10.3

Week 10

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Ideas from Stanford CS224N 2025 L13 (parametric vs open-book). RAFT is the training twist. Original slides; that course is not copied.

Lecture plan

- 1 What RAG is, in one page
- 2 Train the reader on the test stack
- 3 Gold plus distractors
- 4 Weights versus an index

Key goal: if gold is in the prompt and the answer is still the old salary, that is a reader bug.

First time: what / why / formula

- What / why. Train the reader with retrieved snippets in context (gold + distractors) so it learns to cite or ignore, not only recite pretrain weights. Vanilla RAG at test time with a frozen SFT model often ignores the snippet. Index edits then do nothing.
- Architecture. Same retrieve-then-generate stack as RAG. The difference is the training rows and that θ changes. Loss is SFT on the answer given the bundle.
- Formula. $\mathcal{L} = - \sum_{t \in \text{answer}} \log p_{\theta}(y_t \mid y_{<t}, q, \hat{z}_{1:k})$ with $\hat{z}$ containing distractors on purpose.
- Tradeoffs. + Reader learns to use the open book. – Still SFT (can copy); needs a retriever at train time; not a new architecture; gold-only rows never practice skip.

How. Build rows: question + top- k texts (some irrelevant) + target answer that uses gold and skips junk. Lab 10's overlap retriever is the toy ranker.

What RAG is (you need this before RAFT)

- Look up passages from a collection you control, put them in the prompt, generate from them.
- The generator is usually frozen. Edit a PDF and re-index; do not retrain.
- Week 14.1 is the full first-time lecture. This hour trains the reader.

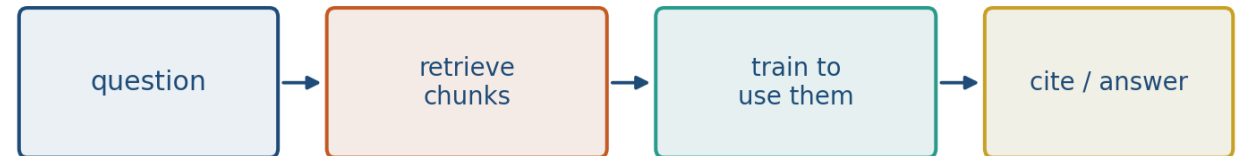
RAG: look up passages, then write the answer from them



SFT on the bundle

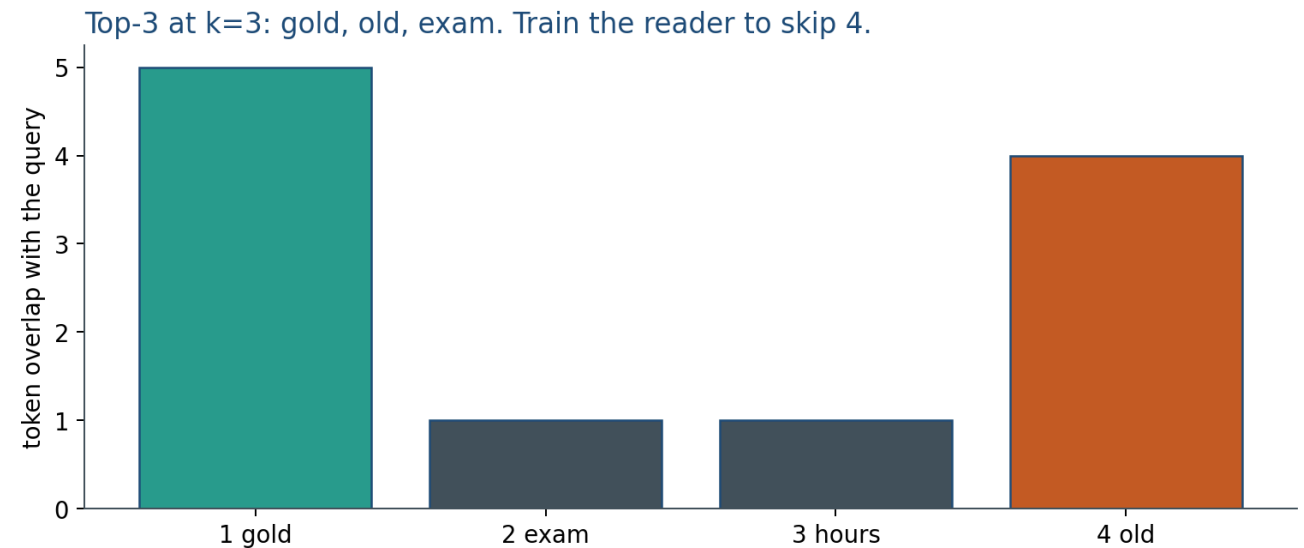
- Parametric recall versus open-book.
- RAFT: question + snippets + an answer that cites or ignores.
- Still SFT. The labels are answers given the bundle.

RAFT: retrieval is in the training loop, not only at demo time



A late-policy question

- Overlap 5, 1, 1, 4. Top-3: gold, old, exam.
- Target: cite snippet 1, skip snippet 4.
- If you only ever trained with gold alone, junk wins.



Two stores

Parametric

Weights. Fast, always on, hard to update. ROME and SFT live here.

Non-parametric

A PDF you can edit, then re-index. Prefer this if the number moves every month.

Whose bug was that

Retrieval

Gold ranked 3rd and $k=2$. The reader never saw it.
Raise k or fix chunking.

Reader

Gold is in the prompt; the model recites the pretrain snapshot. That is RAFT's job. Week 14 is the full stack.

What to take from this lecture

Distractors

Gold-only training never practices skip.

Two writes

A salary in a PDF is not a ROME write.

Lab 10

Token-overlap retrieve, then overwrite one dict key.